Where does the melody go? Measuring and steering melody and lyrics inside an open song model

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Abstract

A person who wants a song model to sing their melody with their words has few ways to check whether the melody survives inside it. We study ACE-Step 1.5, an open model that plans a song as 5 Hz tokens and renders it with a flow-matching transformer, and ask where a given melody is lost and whether it can be restored, with the words, without training. Rendered from tokens computed from their own audio, 16 synthetic melodies keep a pitch-class accuracy of 0.29 (input 0.90), and as dense latents held for all 8 steps, 0.91. A linear probe reads pitch class at 0.58 to 0.67 from the tokens and 0.92 from audio latents, so the loss lies partly in the tokenizer and partly in the renderer's use of them. In an exploratory sweep, a wordless melody guide traded words for notes. A preregistered motif study of repainted flute lines supports literal repetition driven by the pattern, not transposition, sequence extension or analogy. A "lyric clock", an attention bias that ties each word to its note, did not fix timing alone. With a melody guide, 14 of 24 new melodies had half the notes and words right (best of 4 draws, words by Whisper), against 8 without the clock, a gain the registered test did not credit; a registered replication credited the clock (47 against 17 of 96) but missed its capability bar by one melody. A second recogniser does not confirm the word gain; blind listening tests are under way.

Keywords: singing voice generation, melody control, song generation, attention steering, preregistration

1. Introduction

Open song models now write a full song from a lyric and a short text prompt (Yuan et al., 2026; Ning et al., 2025; Yang et al., 2025; Lei et al., 2025; Gong et al., 2026). A person who already has a melody wants something narrower: the model should sing that melody, with those words. This project began with that request failing. The song sounded good, but the melody was not the one that was written, and when the chorus was asked to come back later in the song, it usually did not.

Melody control is well studied for instrumental text-to-music models, where a chroma or pitch curve conditions generation (Copet et al., 2023; Wu et al., 2024; Tal et al., 2024; Tsai et al., 2025), and for singing voice synthesis, which takes notes and lyrics together as a score (Liu et al., 2022; Zhang et al., 2023). SongEcho trains melody conditioning into an earlier version of ACE-Step for cover songs (Li et al., 2026), and MPEcho adds phoneme conditioning to it because melody conditioning alone left many lyric errors (Lee et al., 2026). Our question comes earlier. In a song model that first writes a compact plan and then renders audio from it, where does a given melody go, and can it be put back at inference time, without training, together with the words?

We study one open model, ACE-Step 1.5 (Gong et al., 2026). We test the measuring tools on audio whose notes are exactly known before we test the model. The main studies were preregistered: hypotheses, thresholds and scoring code were committed and hashed before the data existed, each change was recorded in a dated addendum, and every verdict below is reported as the registered rule gives it, failures included. The work makes four contributions.

- A path-by-path measurement of melody loss in ACE-Step 1.5, with a linear probe of the plan tokens (Section 4).
- Hooks on every denoising step of the renderer that read or change its estimate of the clean latent, swap its conditioning or bias its cross-attention, and that leave the output bit-identical when they change nothing (Section 3).
- A preregistered benchmark of motif development with exact symbolic ground truth: repetition, transposition, sequence and analogy, each scored

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against a matched foil (Section 6).
- A "lyric clock": while the renderer produces note j, it pushes the attention heads that ACE-Step uses to align lyrics towards word j, so each word should start on its own note. Three registered tests, B1, B2 and Run C, try it alone and with a melody guide (Section 7).

Code, result files and registrations are public, and a supplement gives the full account of every study and every deviation.

The results are mixed. Through plan tokens, a melody keeps under a third of its pitch classes. Guiding the renderer with the melody gets the notes right or keeps the words, rarely both. The model repeats a phrase when the pattern calls for it but does not transpose it, extend a sequence or complete an analogy. The lyric clock alone did not fix timing; with a guide it helped by Whisper's count, credited by the larger of two registered tests, which missed its capability bar by one melody. A second recogniser does not confirm the word gain, and no listening result is in yet.

2. Related work

Song models and melody control. Recent open song models produce full songs with vocals from lyrics and a style prompt, either autoregressively (Yuan et al., 2026), with latent diffusion (Ning et al., 2025; Gong et al., 2025), or with a language model followed by a separate decoder or renderer, as in ACE-Step 1.5 (Lei et al., 2025; Gong et al., 2026); SongBloom interleaves autoregressive sketching with diffusion refinement (Yang et al., 2025). For instrumental music, melody can also be imposed at inference by optimising the initial noise (Novack et al., 2024). Text-to-song work first synthesises the vocal from a score and lyrics and then generates the accompaniment (Hong et al., 2024), and symbolic systems compose melody and lyrics together (Ding et al., 2025). YingMusic-Singer reports a trade-off between lyric accuracy and melody adherence and reduces it with reinforcement learning (Hao et al., 2026), a trained counterpart of the problem in Section 5; our interventions use no training. Long-range repetition has been studied mostly in symbolic models (Huang et al., 2019); our motif study tests it in an audio model, against exact symbolic answers.

Attention, alignment and steering. Word timestamps can be read from a few cross-attention heads of Whisper (Yeh et al., 2025). ACE-Step 1.5 applies the idea to lyrics: its report describes a reinforcement-learning reward computed by dynamic time warping over the renderer's lyric cross-attention maps (Gong et al., 2026), and its released code reads lyric timestamps from seven fixed cross-attention heads. Adding a fixed bias to attention logits is the mechanism of ALiBi (Press et al., 2022); the lyric clock applies a bias of this kind to the lyric heads during sampling. Linear probes (Alain and Bengio, 2016) have shown what music models encode (Wei et al., 2024), and activation steering has been applied to music generators (Facchiano et al., 2025; Staniszewski et al., 2026; Singh et al., 2026). In ACE-Step v1.0, activation patching places the control of concepts such as vocals and instruments in cross-attention layers 7 and 8 of 24 (Staniszewski et al., 2026); the lyric heads used here lie in layers 2 to 6 of ACE-Step 1.5's renderer, counted from 0 as in its code.

Measuring singing. Lyrics transcription of sung audio is harder than speech recognition (Ou et al., 2022), and Whisper is known to produce fluent text for non-speech audio (Koenecke et al., 2024; Barański et al., 2025). We follow the basic form of preregistration argued for in NLP (van Miltenburg et al., 2021), and confidence intervals come from bootstraps that resample melodies (Field and Welsh, 2007).

3. The model, the hooks and the measures

3.1 ACE-Step 1.5 as used here

ACE-Step 1.5 has two parts. A language model, which we call the planner, writes a plan of the song as discrete tokens at 5 per second; the tokens are finite scalar quantization codes (Mentzer et al., 2024), which our probe reads as six digits with 8, 8, 8, 5, 5 and 5 levels. A flow-matching transformer, the renderer, turns the plan, the lyric and a caption into latents of a variational autoencoder (VAE) at 25 frames per second with 64 channels, which the VAE decodes to 48 kHz audio. The renderer has 384 cross-attention heads (16 in each of 24 layers), and each of its attention queries covers two latent frames (80 ms). We use the "turbo" checkpoint, which samples in 8 Euler steps with the time shift of 3.0 that the ACE-Step 1.5 report gives for the 8-step model (Gong et al., 2026).

Three of the model's tasks appear in this paper. Text-to-music generates from the caption and the lyric, optionally with given plan tokens. Cover generation gives the renderer the dense 25 Hz latents of a source recording as context; with a strength of $k/8$, the context is held for the first k of the 8 steps, after which text and lyric conditioning take over. Repaint regenerates one time window of a source and keeps the rest. All GPU runs used one NVIDIA L4, bf16 weights and no planner language model. The first GPU run (Sections 4 and 5) recorded neither the code commit nor the weights revision; the later runs pinned ACE-Step to commit `ca1e85f` and its weights to a fixed revision.

3.2 Hooks on every denoising step

The turbo sampler integrates

$$x_t = t\,\epsilon + (1-t)\,x_0, \qquad v = \epsilon - x_0, \tag{1}$$

so at every call of the renderer its estimate of the clean latent is $\hat{x}_0 = x_t - t\,v$. Our wrapper around the ren-

derer's forward pass reads $\hat{x}_0$ at each step and can replace it with an edited x'_0 by returning $v' = (x_t - x'_0)/t$. The same wrapper can swap the conditioning before a call, for example the context latents part way through sampling. A bias hook adds a fixed matrix to the additive attention mask of chosen heads before the softmax; the lyric clock is built on it.

Before every GPU study we checked that a hook doing nothing changes nothing: an identity edit, and the bias hook at zero bias, left the output bit-identical to the unhooked run on CPU and GPU (maximum difference 0.0). With deterministic kernels, a repeated run and a run in a fresh process gave identical latents and audio. The same seed gives different audio on CPU and GPU, so every confirmatory run was made on one GPU type.

3.3 Measures

Before any model run we measured the tools on audio where every note is known, rendered with FluidSynth (FluidSynth developers, 2023) and the FluidR3_GM soundfont (Wen, 2008). The pitch trackers are accurate on a solo line and much less so under a loud band: the raw pitch accuracy of SwiftF0 (Nieradzik, 2025) on a solo piano lead is 0.947 and falls to 0.095 when the band plays at MIDI velocity 90. Two rules follow. A score computed on a full mix is always read next to the same score on the input mix or against a chance level from other melodies, and singing is scored only in clips whose caption asks for no instruments.

Pitch-class accuracy. For every frame where the written melody sounds, the dominant chroma bin of the output is compared with the written pitch class, octave ignored, at the best single time lag within ±0.3 s. Scoring each output against the other melodies at the same tempo gives the chance level.

Note pitch accuracy (NPA). A written note counts as sung right if the median SwiftF0 frequency over the central 60% of the note lies within 50 cents of the written pitch, in the written octave, after one global lag search within ±0.5 s. NPA_v is NPA on the vocal stem separated by Hybrid Transformer Demucs (Rouard et al., 2023); its chance level is about 0.10.

Word recall (WR). Whisper large-v3 (Radford et al., 2023), at a pinned revision with greedy decoding, transcribes the full mix without being given the lyric. Lyric words are matched to Whisper's words in order by exact normalised spelling, and WR is the share of lyric words matched. We report recall rather than word error rate because Whisper writes text for wordless audio: in one sweep, 106 of 504 transcripts had a word error rate above 1. A clip is *wordless* if its transcript is empty, at least half vowel sounds, or only known hallucination strings and vowel sounds. A wav2vec 2.0 model trained with CTC (Baevski et al., 2020), decoded with the same matcher, gives a second opinion; it was trained on read English speech and was not checked on singing.

Joint score. A clip passes the vocal gate if its separated vocal stem holds at least 20% of the stem energy and at least 1.0 s of its frames reach −40 dBFS. A clip's joint score is $q = \mathrm{gate} \times \min(\mathrm{NPA}_v, \mathrm{WR})$, and the clip is *joint* if $q \geq 0.5$: it has vocals and at least half of the notes and half of the words are right. NPA and WR are scored separately, so a joint clip need not sing the recognised words on the right notes. No score here, and none of the registered listening questions, says whether a voice sounds natural.

4. Where the melody is lost

Each path through the model. We rendered 16 random eight-bar melodies, half with a piano lead and half with a flute lead, over block chords, bass and drums, at 80, 120 and 160 BPM. Each clip went through the model by several paths with one seed: a VAE round trip; the dense latents as context for all 8 steps; the plan tokens that the model's tokenizer computes from the clip's own audio; the same tokens with a caption that asks for different instruments; and text-to-music with no audio condition. The token path is an easy case for the planner, since the tokens come from the target audio itself.

With the source latents held for all 8 steps, the output keeps the input's pitch classes; through the plan tokens it does not (Figure 1). Pooled over the three tempos, with 95% bootstrap intervals over the 16 melodies, pitch-class accuracy is 0.90 [0.84, 0.94] for the input itself, 0.91 [0.86, 0.95] with dense context, 0.90 after a VAE round trip, 0.29 [0.25, 0.34] through the plan tokens, 0.31 with new instrumentation and 0.10 with no audio condition. Chance from mismatched melodies is 0.19 for the input and 0.10 for the token path; after subtracting it, the token path keeps 28% [22%, 34%] of the melody information that the input carries. We found no large tempo effect. The dense path is close to a reconstruction of the input, so it shows that the renderer can pass a melody through, not that it keeps one while it also generates new content. An earlier CPU run with 4 melodies per tempo gave the token path 0.51 at 90 BPM; the larger GPU run did not reproduce it.

What the plan tokens hold. To locate the loss, we trained linear probes (Alain and Bengio, 2016) to read the pitch class of the lead from each 200 ms frame of 48 four-bar melodies at 120 BPM, each with a piano, flute and violin lead. Probes learned on 42 melodies and were tested on the other 6. From the plan tokens the probe reaches 0.575, and 0.667 with one-hot features of the six code digits; from the audio latents pooled

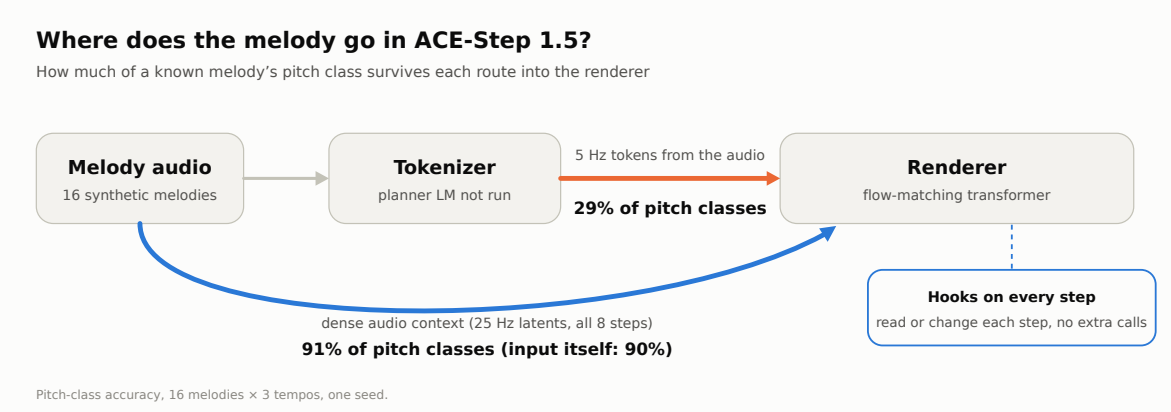


Figure 1: The two routes into the renderer and how much of a known melody's pitch class survives each (16 melodies × 3 tempos, one seed). The dense route holds the source latents for all 8 steps, a cover at full strength.

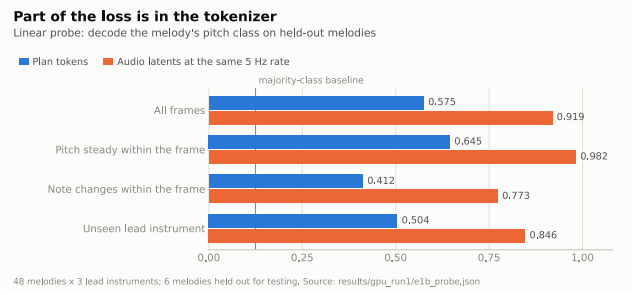


Figure 2: Linear-probe pitch-class accuracy on 6 held-out melodies (720 frames), plan tokens against audio latents pooled to the same 5 Hz rate. One split; no confidence intervals, as the token file was not saved.

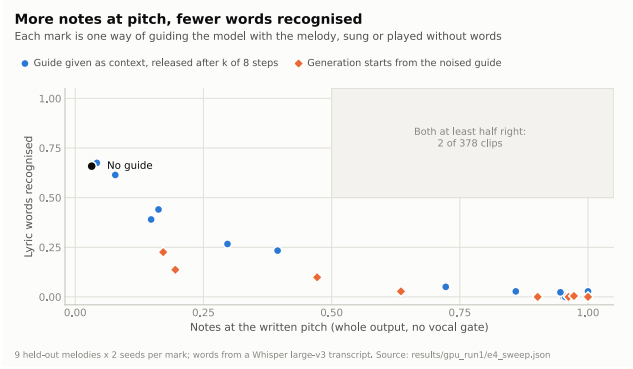


Figure 3: Notes versus words: each mark is one guide setting, averaged over 9 held-out melodies × 2 seeds; notes at the written pitch in the whole output against lyric words recognised by Whisper.

to the same 5 Hz rate it reaches 0.919; the majority class gives 0.125 (Figure 2). The low frame rate is not the whole reason: on frames where the note does not change, the latent probe reaches 0.982 and the token probe 0.645. A linear probe only gives a lower bound on what the tokens hold, and the probe used other stimuli than the path test. Still, it reads pitch class from the tokens well above the 0.29 that the token path renders, so part of the loss is in the tokenizer and part happens when the renderer uses the tokens. The tokens also mix the instrument into the plan: changing only the lead instrument keeps about 27% of tokens identical, about as few as changing the melody (29%); re-encoding the same audio keeps 85%, because the encoder samples.

Pitch in the renderer's latents (exploratory). A classifier trained on chromatic scales in 24 General MIDI instruments and tested on 6 others reads pitch among 32 classes at 0.879 from the 64-dimensional VAE latent (chance 0.031). Since the melody survives in the renderer's dense input and latents but not through the plan, the rest of the project built control inside the renderer.

5. Notes versus words (exploratory)

We asked whether the renderer can sing a given melody with given words when it hears the melody without words. The items are 12 English lyric lines of 11 to 13 one-syllable words, one note per word, with random melodies at 100 BPM and an a cappella caption. Three items were used to try settings and nine were held out. The guide is the melody rendered by FluidSynth as Voice Oohs or as a clarinet, given as dense context for the first k of 8 steps ($k \in \{1,2,3,4,6,8\}$) or, as in SDEdit (Meng et al., 2022), as a noised starting latent at four noise levels. With a no-guide reference that makes 21 settings, each run with 2 seeds.

The result is a trade-off with almost nothing in the corner where both are right (Figure 3). Without a guide the model does not know the melody: 0.03 of the notes are at the written pitch, the same as the chance level from mismatched melodies (0.04), while 0.66 of the words are recognised. Every setting that put 0.85 or more of the notes at pitch kept at most 0.03 of the

Table 1: Motif conditions. The gap follows the three context units.

Condition	Context	Answer asked for
lit	A, B, A	B
ctl	A, B, C	none (lag control)
rel+k, $k \in \{+2, +5, -3\}$	A, B, T_kA	T_kB
seq	A, T_2A, T_4A	T_6A
ana	A, T_5A, B	T_5B
null	silence	none

words. Between the extremes the two move against each other: the Voice Oohs guide held for 3 steps gave 0.40 of the notes and 0.23 of the words, and for 4 steps 0.86 and 0.03. Only 2 of the 378 held-out clips had at least half of the notes and half of the words right. Because this sweep used no source separation, a high NPA at large k may come from the wordless guide carried through rather than from singing.

A working hypothesis, the timing-conflict hypothesis, came out of these outputs. Without a guide the model sings these lines early and fast. Under a moderate guide, the guide's note times and the model's own lyric timing disagree, and the model resolves the conflict by dropping the words. If the lyric attention could be moved onto the guide's note times, the two signals would agree; that is the idea behind the lyric clock (Section 7). A simpler rival explanation fits equally well: the longer a wordless input is held, the closer the output stays to it, and that input has no words.

6. Motif development (preregistered)

6.1 Design

Songs return to earlier material, often changed: a phrase comes back, comes back higher, or continues a pattern. We ask which of these an audio song model does when it fills a gap. Each item is a context of three 4 s units (0 to 12 s), a gap from 12 to 16 s that the model fills by repaint, and a unit D after the gap. The units are short random melodies rendered as a solo flute line. Table 1 lists the conditions. In lit the pattern A B A asks for B. In ctl, A B C, nothing asks for B, but B sits exactly as far before the gap (8 s) as in lit, so the contrast separates a repeat driven by the pattern from a repeat of whatever was heard 8 s earlier. T_k transposes by k semitones.

Every possible answer is rendered exactly from symbols, together with a foil for each answer: a reordering of its pitches, with the same rhythm, in which no note keeps its place. A gap's accuracy against an answer is the share of the answer's notes matched by SwiftF0 within 50 cents, and each index is the accuracy against the answer minus the accuracy against its foil, so 0 is chance. CPI is this copy index for B; RI$_k$ uses T_kB and LI$_k$ the unmoved B; ANA uses T_5B. H4 uses the share of seq gaps that a classifier calibrated on renders assigns to T_6A. Before any model run, the scorer had to sort renders of every answer correctly at least 95% of the time and detect one wrong note with an AUC of at least 0.9. A gap passes the quality gate if at least 60% of it is voiced, its Basic Pitch (Bittner et al., 2022) polyphony is at most 1.3, and its level is within 6 dB of the context. The hypotheses and thresholds are in Table 2; the registered priors were about 0.65 for H1 and 0.1 to 0.2 for H3 to H5.

A failed first run. The first confirmatory run failed its registered quality gate: only 59 of 120 non-null gaps passed, where 75% was required, so by the registered rule it supports no claim, positive or negative. The diagnosis found that every repaint had received the text-to-music instruction instead of the repaint instruction, and that within a seed every item had started from the same noise. A dated addendum registered one retry before any of its generations, with 36 new melodies, 4 draws per melody with their own seeds, the repaint instruction and a pilot that chose the instrument. The addendum kept H1 to H5 and every threshold; it was written after the first run's indices had been computed and recorded, and its sample size came from a simulation calibrated on those outcomes. Before the retry, the scorer check failed because some notes of B fell outside the scored window; the melodies were rebuilt under the registered remedy and the check then passed in all three timbres. The confirmatory grid ran 36 melodies × 8 conditions × 4 draws = 1152 repaints, with none missing, and 804 of 1008 non-null gaps passed the quality gate (756 needed).

6.2 Results

Repetition: H1 and H2 supported. After A B A the gap matches B with CPI 0.471; in absolute terms it matches 0.66 of B's notes, against 0.07 in null gaps. After A B C it matches B far less, 0.106, and the difference, 0.365, clears the H2 threshold. Lit gaps passed the quality gate somewhat more often than ctl gaps (0.875 against 0.778), but the verdict uses all draws, so the gate does not select the H2 population, and H2 also holds in the gate-passing draws (0.403), as the rule requires.

Development: H3 to H5 not supported. After A B T_kA the gap does not move B by k: RI$_k$ is close to 0 for every k. The gap matches the unmoved B (LI$_k$ 0.10 to 0.13) at about the level of the lag copy in ctl. In seq the gap was never classified as the next step T_6A (0 of 144 draws), and after A T_5A B it does not match T_5B.

So the repeat follows the pattern, as the ctl contrast shows, but the model does not carry a transposition over to B, extend a sequence or complete an analogy at the registered thresholds. These are pitch-tracker scores of flute lines filled by repaint.

Table 2: Motif study, registered indices over all draws (36 melodies, 144 draws per condition), mean and 95% cluster-bootstrap interval over melodies. H3 is not supported.

	Index	Value [95% interval]	Threshold	Verdict
H1	CPI(lit)	0.471 [0.405, 0.536]	≥ 0.25	supported
H2	CPI(lit) − CPI(ctl)	0.365 [0.306, 0.425]	≥ 0.15	supported
H3	RI_{+2}	−0.021 [−0.043, 0.001]	≥ 0.15	not met
	RI_{+5}	0.002 [−0.024, 0.028]	≥ 0.15	not met
	RI_{-3}	0.020 [−0.002, 0.041]	≥ 0.15	not met
H4	T_6A in seq	0 of 144	≥ 20%	not supported
H5	ANA	0.008 [−0.018, 0.034]	≥ 0.15	not supported
	CPI(ctl)	0.106 [0.063, 0.152]		
	$LI_{+2/+5/-3}$	0.131 / 0.098 / 0.103		

7. The lyric clock

7.1 Do the lyric heads follow the words?

Before building on the seven lyric heads of Section 2, a read-out, registered as exploratory and as the gate for B1, asked whether they give the onset of each sung word, using 88 a cappella clips and Whisper's word time stamps as the reference. On the 50 evaluable clips, the median onset error was 0.21 s, and the alignment heads beat the median of 200 random sets of 7 heads in 49 of 50 clips (random sets: 1.95 s). Two cautions limit what 0.21 s means. Whisper and a forced aligner (Pratap et al., 2024) differ by a median of 0.39 s on these clips, so the error is within the uncertainty of the reference. And the ACE-Step 1.5 report's training diagram ends the turbo model's training with a reinforcement-learning stage, with the attention-alignment reward of Section 2; if that reward shaped these heads, heads that follow word timing are expected, so beating random heads is a sanity check rather than a finding. The read-out shows that the heads follow word timing in finished audio; it does not show that they control it.

7.2 The clock

The lyric clock adds a fixed value β to the attention scores (before the softmax) of the seven heads at all 8 steps, wherever an audio frame whose centre lies inside word j's window attends to word j's lyric tokens. Word j is sung on note j, and its window runs from that note's written onset to the next note's onset. All other scores are untouched. If a head gives a word attention probability p inside its window, the bias raises it to $pe^{\beta}/(pe^{\beta}+1-p)$: from $p = 0.10$ to 0.45 at $\beta = 2$ and 0.997 at $\beta = 8$.

The clock alone (B1, not supported). Without a guide, on the 12 lines of Section 5 with 3 draws each, the model sang the words a median of 3.62 s before their written notes. The registered rule asked for at least a 30% reduction of that timing error in 9 of 12 lines at $\beta = 2$; 2 lines improved, so B1 was not supported. At $\beta = 4$, outside the rule, 6 lines improved, with word recall down from 0.59 to 0.52 and the words still 2.92 s early.

7.3 The clock with a melody guide (B2 and Run C)

Design. B2 combines the clock with the guides of Section 5. Its claim concerns a pipeline that a user could run: one recipe fixed on development lines, the user's melody rendered as a wordless guide, the user's lyric, 4 draws (generations with their own seeds), and an automatic selector that keeps the draw with the largest q, which needs only the melody and the lyric. A registered function chose the recipe on the 12 development lines from the 25 of 30 registered configurations left after a registered time-budget rule: the Voice Oohs guide held for the first 3 of 8 steps with the clock at $\beta = 8$ (S^*). The test ran on 24 new lines and melodies against controls that share S^*'s guide and seeds: the same guide without the clock, the clock on a random set of 7 heads, and the clock with the windows rotated by half a line so that each word gets another word's note. A reference arm held the guide for 4 steps without the clock. The outcome is J, the share of melodies whose kept draw is joint. The rule credits the clock if ΔJ, the gain in J over the best unclocked guided arm, is at least 0.20 and its bootstrap 2.5th percentile is above 0, and calls the capability reached if $J(\text{clock}) \geq 0.50$.

B2: capability reached, clock not credited. The kept draw of the clocked recipe was joint on 14 of 24 melodies, against 8 for the guide alone (Table 3). $\Delta J = 0.25$ cleared the size bar, but the bootstrap's 2.5th percentile was exactly 0, so the rule did not credit the clock. The test had little power at this effect size: in the registered simulation it passed with probability 0.27 at an effect of about 0.21 and 0.65 at about 0.34.

Run C: clock helps, capability not reached. Because 24 melodies were too few to settle the clock's effect, Run C tested the two B2 recipes again on 96 new lines and melodies, fixed in advance with no selection stage. Its rule is looser than B2's: the clock is credited if the bootstrap 2.5th percentile of ΔJ is above 0, with no size bar and the guide alone as the only comparator, and the capability bar is 48 of 96 melodies. The clocked recipe was joint on 47 melodies and the guide alone on 17: $\Delta J = 0.31$, 95% interval 0.198 to 0.427, so the clock was credited, and ΔJ would also have met B2's

Table 3: The lyric clock with a melody guide. B2: 24 new melodies; Run C: 96 further melodies; 4 draws per melody and arm. All guided arms use the Voice Oohs guide for 3 of 8 steps except the last row. J: melodies whose kept draw (best of 4 by q) is joint. Kept q: mean q of the kept draws. NPA$_v$, WR and Wordless: means over all clips. CTC J: melodies whose kept draw is joint when words come from the CTC recogniser on the mix. Onset: median word-start error in seconds over clips with at least 4 matched words. Result-file names of the arms: S*, S*_b0, S*_rand, S*_shuf.

	J		Kept q		NPA$_v$		WR		Wordless		CTC J		Onset (s)	
Arm	B2	C	B2	C	B2	C	B2	C	B2	C	B2	C	B2	C
Guide + clock (S*)	14	47	0.517	0.465	0.44	0.43	0.50	0.44	0.02	0.04	2	9	0.42	0.40
Guide alone	8	17	0.385	0.281	0.45	0.41	0.32	0.23	0.28	0.41	3	9	1.73	2.17
Clock on random heads	8	20	0.353	0.286	0.44	0.41	0.28	0.19	0.31	0.42	1	3	1.86	2.02
Clock, rotated windows	7	9	0.391	0.312	0.41	0.37	0.31	0.26	0.03	0.09	2	1	4.49	4.79
Guide for 4 steps, no clock	3	–	0.175	–	0.75	–	0.07	–	0.71	–	1	–	3.75	–

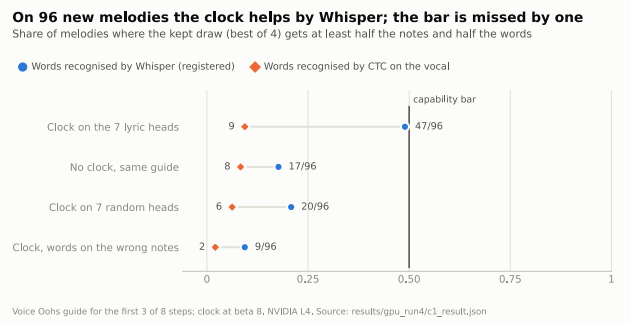


Figure 4: Run C: share of 96 new melodies whose kept draw (best of 4) has at least half the notes and half the words right. Blue: words by Whisper, as registered. Orange: words by the CTC recogniser on the separated vocal, on the same kept draws. The vertical line is the registered capability bar.

size bar; the capability bar was missed by one melody. In kept q, random heads and rotated windows score about as low as the guide alone (0.29 and 0.31 against 0.28), which points to the lyric heads and to the note timing. With the clock the words start a median of 0.40 s from their notes, against 2.17 s without it, over different clips (269 and 125); on B2's 39 seed-paired clips the figures are 0.40 s and 1.72 s.

7.4 What limits the result

The words rest on Whisper. The second recogniser does not confirm them (Figure 4). With CTC word recall on the same kept draws, the clocked recipe and the guide alone tie in Run C on the mix, 9 melodies each (9 against 8 on the separated vocal), and in B2 they score 2 and 3. Of the 47 melodies joint by Whisper with the clock, 9 (19%) are also joint by CTC on the vocal, against 8 of 17 (47%) without it. Much of Whisper's gain is the clock turning wordless output into words: 41% of the guide alone's clips are wordless in Run C, against 4% with the clock. The rotated-window clock also removes most wordless outputs, so part of the clock's effect on words does not depend on its timing; yet it is joint on fewer melodies than the guide alone, partly because the in-order matcher does not count words sung out of order (in B2, matching in any order lifts it from 7 to 10 melodies). Scored against the other lyrics of B2, Whisper's recall is 0.14 with the clock and 0.09 without, so the gain above chance is 0.12 rather than 0.17, and no clip reaches 0.5 against another lyric, so J is not inflated by chance. Because the kept draw is chosen by Whisper, a second recogniser scoring the same draws is biased against the arm that gains most from selection, and the CTC model reads every arm poorly. Whether the words are clear remains open.

Selection and margins. The kept draw is the best of 4 by the score that decides it. With a single draw the clock's gain in J is 0.19 in B2 and 0.16 in Run C, so the gap is there without selection. Both capability verdicts sit inside the noise. B2 reached its bar by two melodies: one correct note fewer on every clip would leave it exactly on the bar, and two fewer at 10 of 24. In Run C, J for the clocked recipe has a 95% interval of 0.39 to 0.59, and one correct note more or fewer on every clip would move its count to 54 or 35. Both recipes scored lower in Run C than in B2, and Run C's gain cannot be told apart from B2's (difference 0.06, 95% interval −0.20 to 0.32).

8. Discussion and limitations

Where the melody goes. In ACE-Step 1.5 a melody that has to pass through the 5 Hz tokens arrives in the output mostly lost, even when the tokens are computed from the melody's own audio. For a user who brings a melody, this places control in the renderer or in a better plan, not in the caption. We did not test the planner generating tokens from a symbolic melody, nor optimised tokens, so this does not bound what a better plan could carry.

Notes and words. The clock moved words closer to their notes, but the evidence that it made them clearer rests on Whisper alone (Section 7.4). With the guide length fixed, the clock raised the words, which argues

against the simpler rival of Section 5; but the rotated-window clock also removed wordless output, which points to a third reading: attention forced onto the lyric tokens at all, whatever the timing.

Motifs. For a user who relies on repaint, a phrase can be brought back but not developed; the test used short instrumental phrases only.

Limitations. Every result concerns one model with the 8-step turbo sampler; other models and samplers may behave differently. All runs bypass the planner language model; the clock was not tested with planner tokens. All melodies come from random generators rendered by FluidSynth, with no real singers and no real songs. The sung lines are English, one syllable per note, at 100 BPM, with one a cappella caption; melismas, other languages and accompaniment were not tested. Pitch rests on trackers and words rest on Whisper, and singing quality and naturalness are not measured at all. Samples are small: 36 melodies in the motif study, 24 in B2 and 96 in Run C. A comparison with a commercial song model, registered before any generation, has not been run. The custody of the registrations and every deviation are in the supplement; four bear on interpretation. The first registration and its first addendum were committed privately, so their public time stamps rest on addendum 2, committed before the corrected motif run. B2's time-budget rule acted on a projection about six times too long, so it also removed the no-guide reference arm. Two later sessions reopened Run C's finished state, and its result package was re-zipped by hand with no stored ledger hash; every kept draw and J recomputed from the ledger rows matches. The first motif run's pilot did not follow the registered caption rule.

What would change these conclusions. Listeners who do not hear the words that Whisper counts, or who do not hear the repeated phrase that the tracker scores, would remove the corresponding claims. Blind listening tests of the B2 and Run C pairs, registered before any rating, ask whether people hear the melody and the words better with the clock; their ratings close on 2026-11-30, and their results will be reported in a later version.

Reproducibility

Code, result files, registrations and figures are at `https://github.com/ninondev/DaCapo` under the MIT license; this version describes commit `9ddcdbc`. Every number in this paper comes from, or is computed by the repository's code from, a file in `results/` or a registration file; the pooled intervals of Section 4 are recomputed from the per-clip rows (`LOG.md`). A script draws every figure from the result files. The supplement lists the registration files with their time stamps and SHA-256 hashes, the pinned model and tool versions, and every deviation. Audio, latents and the blind listening keys are kept outside the repository.

Author contributions

The author led the project. The author set the research questions; searched for a model to study and chose ACE-Step 1.5 with help from a general-purpose AI assistant; decided to build melody control inside the renderer; required every main study to be preregistered, so that its verdict follows rules fixed before the data existed, and the claims about how the outputs sound to rest on blind listening; ran every GPU experiment; and made the decisions that the registrations record as the author's, such as the remedy after the failed scorer check. Within the questions and rules set by the author, AI coding tools designed the experiments, wrote the registrations and the code, ran the CPU checks, analysed the results and drafted this article and its supplement, and the author reviewed their work. The author takes responsibility for the content of this article.

Competing interests

The author has no competing interests to declare.

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References

Alain, G. and Bengio, Y. (2016). Understanding intermediate layers using linear classifier probes. arXiv:1610.01644.

Baevski, A., Zhou, Y., Mohamed, A., and Auli, M. (2020). wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems*, volume 33.

Barański, M., Jasiński, J., Bartolewska, J., Kacprzak, S., Witkowski, M., and Kowalczyk, K. (2025). Investigation of Whisper ASR hallucinations induced by non-speech audio. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5.

Bittner, R. M., Bosch, J. J., Rubinstein, D., Meseguer-Brocal, G., and Ewert, S. (2022). A lightweight instrument-agnostic model for polyphonic note transcription and multipitch estimation. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 781–785.

Copet, J., Kreuk, F., Gat, I., Remez, T., Kant, D., Synnaeve, G., Adi, Y., and Défossez, A. (2023). Simple and controllable music generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

Ding, S., Liu, Z., Dong, X., Zhang, P., Qian, R., Huang, J., He, C., Lin, D., and Wang, J. (2025). SongComposer: A large language model for lyric and melody generation in song composition. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7108–7127.

Facchiano, S., Strano, G., Crisostomi, D., Tallini, I., Mencattini, T., Galasso, F., and Rodolà, E. (2025). Activation patching for interpretable steering in music generation. arXiv:2504.04479.

Field, C. A. and Welsh, A. H. (2007). Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390.

FluidSynth developers (2023). FluidSynth: A SoundFont synthesizer. `https://www.fluidsynth.org/`. Version 2.3.4, LGPL-2.1-or-later.

Gong, J., Song, Y., Zhao, W., Wang, S., Xu, S., Guo, J., and Yang, X. (2026). ACE-Step 1.5: Pushing the boundaries of open-source music generation. *arXiv preprint arXiv:2602.00744*.

Gong, J., Zhao, S., Wang, S., Xu, S., and Guo, J. (2025). ACE-Step: A step towards music generation foundation model. *arXiv preprint arXiv:2506.00045*.

Hao, C., Zheng, J., Ma, G., Jiang, Y., Chen, H., Tian, W., Chen, G., Chen, Z., and Xie, L. (2026). YingMusic-Singer: Controllable singing voice synthesis with flexible lyric manipulation and annotation-free melody guidance. In *Proc. Interspeech 2026*, pages 1576–1580.

Hong, Z., Huang, R., Cheng, X., Wang, Y., Li, R., You, F., Zhao, Z., and Zhang, Z. (2024). Text-to-song: Towards controllable music generation incorporating vocal and accompaniment. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6248–6261.

Huang, C.-Z. A., Vaswani, A., Uszkoreit, J., Simon, I., Hawthorne, C., Shazeer, N., Dai, A. M., Hoffman, M. D., Dinculescu, M., and Eck, D. (2019). Music transformer: Generating music with long-term structure. In *International Conference on Learning Representations (ICLR)*.

Koenecke, A., Choi, A. S. G., Mei, K. X., Schellmann, H., and Sloane, M. (2024). Careless whisper: Speech-to-text hallucination harms. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 1672–1681.

Lee, W.-J., Yeh, H.-Y., Hu, T.-Y., Tan, C.-P., Tsai, F.-D., and Yang, Y.-H. (2026). MPEcho: A melody and phoneme-aware generative framework for controllable cover song generation. *arXiv preprint arXiv:2607.26698*. Accepted to the 27th International Society for Music Information Retrieval Conference (ISMIR 2026).

Lei, S., Xu, Y., Lin, Z., Zhang, H., Tan, W., Chen, H., Yu, J., Zhang, Y., Yang, C., Zhu, H., Wang, S., Wu, Z., and Yu, D. (2025). LeVo: High-quality song generation with multi-preference alignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38.

Li, S., Li, Y., Wang, Z., Zhang, Y., Wu, F., Deussen, O., Lee, T.-Y., and Dong, W. (2026). SongEcho: Towards cover song generation via instance-adaptive element-wise linear modulation. In *International Conference on Learning Representations (ICLR)*.

Liu, J., Li, C., Ren, Y., Chen, F., and Zhao, Z. (2022). DiffSinger: Singing voice synthesis via shallow diffusion mechanism. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 11020–11028.

Meng, C., He, Y., Song, Y., Song, J., Wu, J., Zhu, J.-Y., and Ermon, S. (2022). SDEdit: Guided image synthesis and editing with stochastic differential equations. In *The Tenth International Conference on Learning Representations (ICLR)*.

Mentzer, F., Minnen, D., Agustsson, E., and Tschannen, M. (2024). Finite scalar quantization: VQ-VAE made simple. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Nieradzik, L. (2025). SwiftF0: Fast and accurate monophonic pitch detection. arXiv preprint arXiv:2508.18440.

Ning, Z., Chen, H., Jiang, Y., Hao, C., Ma, G., Wang, S., Yao, J., and Xie, L. (2025). DiffRhythm: Blazingly fast and embarrassingly simple end-to-end full-length song generation with latent diffusion. *arXiv preprint arXiv:2503.01183*.

Novack, Z., McAuley, J., Berg-Kirkpatrick, T., and Bryan, N. J. (2024). DITTO: Diffusion inference-time T-optimization for music generation. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 38426–38447.

Ou, L., Gu, X., and Wang, Y. (2022). Transfer learning of wav2vec 2.0 for automatic lyric transcription. In *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, pages 891–899.

Pratap, V., Tjandra, A., Shi, B., Tomasello, P., Babu, A., Kundu, S., Elkahky, A., Ni, Z., Vyas, A., Fazel-Zarandi, M., Baevski, A., Adi, Y., Zhang, X., Hsu, W.-N., Conneau, A., and Auli, M. (2024). Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25(97):1–52.

Press, O., Smith, N. A., and Lewis, M. (2022). Train short, test long: Attention with linear biases enables input length extrapolation. In *The Tenth International Conference on Learning Representations (ICLR)*.

Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., and Sutskever, I. (2023). Robust

speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR.

Rouard, S., Massa, F., and Défossez, A. (2023). Hybrid transformers for music source separation. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5.

Singh, N., Cherep, M., and Maes, P. (2026). Discovering and steering interpretable concepts in large generative music models. In *The Fourteenth International Conference on Learning Representations (ICLR)*.

Staniszewski, Ł., Zaleska, K., Modrzejewski, M., and Deja, K. (2026). TADA! tuning audio diffusion models through activation steering. arXiv:2602.11910.

Tal, O., Ziv, A., Gat, I., Kreuk, F., and Adi, Y. (2024). Joint audio and symbolic conditioning for temporally controlled text-to-music generation. In *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, pages 264–271.

Tsai, F.-D., Wu, S.-L., Lee, W., Yang, S.-P., Chen, B.-R., Cheng, H.-C., and Yang, Y.-H. (2025). MuseControlLite: Multifunctional music generation with lightweight conditioners. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 60266–60279.

van Miltenburg, E., van der Lee, C., and Krahmer, E. (2021). Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 613–623.

Wei, M., Freeman, M., Donahue, C., and Sun, C. (2024). Do music generation models encode music theory? In *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, pages 680–687.

Wen, F. (2008). FluidR3_GM: Fluid (R3) General MIDI SoundFont. Debian and Ubuntu package `fluid-soundfont-gm` 3.1-5.3, `https://packages.debian.org/fluid-soundfont-gm`. MIT license.

Wu, S.-L., Donahue, C., Watanabe, S., and Bryan, N. J. (2024). Music ControlNet: Multiple time-varying controls for music generation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32:2692–2703.

Yang, C., Wang, S., Chen, H., Tan, W., Yu, J., and Li, H. (2025). SongBloom: Coherent song generation via interleaved autoregressive sketching and diffusion refinement. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38.

Yeh, S.-L., Meng, Y., and Tang, H. (2025). Whisper has an internal word aligner. In *2025 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–7.

Yuan, R., Lin, H., Guo, S., Zhang, G., Pan, J., Zang, Y., Liu, H., Liang, Y., Ma, W., Du, X., et al. (2026). YuE: Scaling open foundation models for long-form music generation. In *International Conference on Learning Representations (ICLR)*.

Zhang, Y., Xue, H., Li, H., Xie, L., Guo, T., Zhang, R., and Gong, C. (2023). VISinger2: High-fidelity end-to-end singing voice synthesis enhanced by digital signal processing synthesizer. In *Proc. Interspeech 2023*, pages 4444–4448.